

DroidLink Periscope Complete Guide

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ESP32-C3 Super Mini controller for the Printed-Droid Periscope Logic Lights.

This guide covers hardware, installation, USB-console first-time setup, Master registration, normal operation, commands, recovery, and troubleshooting. The firmware keeps the original Periscope lighting effects and adds wireless control through the DroidLink Master.

The Periscope can still be controlled directly from USB Serial, but normal operation is through the DroidLink Master.

What you need

- The ESP32-C3 Super Mini Periscope controller and connected logic lights
- The correct 5 V power supply and a common ground for the controller and lights
- A USB data cable
- The DroidLink Installer and its Console Log set to 9600 baud
- The DroidLink Master's MAC address
- One unused DroidLink Device ID from 2 through 13

Controlling the Periscope

The communication path is:

```
Display / RC / Master Sequence
      ↓
DroidLink Master
      ↓
Wireless DroidLink connection
      ↓
DroidLink Periscope
      ↓
Periscope LED Effects
```

The DroidLink command prefix for the Periscope is:

```
:PS
```

Examples:

```
:PSOFF
```

Turns the Periscope lights off.

```
:PSQ1
```

Runs Periscope Sequence 1.

Runs Main LED Effect 1, White, at Speed 5.

Installation and First-Time DroidLink Setup

The Periscope and Master must know each other's MAC addresses.

Step 1 — Install the Periscope Firmware

Install the DroidLink Periscope firmware using the DroidLink Installer.

After installation, wait for the blue setup light, open **Console Log**, and then select **Reset Device** or press the Periscope's physical reset button.

During first-time setup, normal Periscope operation does not start. The firmware only displays the setup instructions and accepts the node ID and Master MAC. Wireless communication, lighting effects, sequences, and normal command processing remain inactive until both values have been saved.

The Periscope will display its device MAC address:

```
DroidLink Periscope -- Boot  
  
Device MAC: XX:XX:XX:XX:XX:XX
```

Your actual MAC address will be different.

Copy this address.

Step 2 — Add the Periscope to the Master

Open Master System Setup. Select **Add Slave**, enter a useful name such as `Periscope`, and enter the exact Periscope Device MAC.

The Periscope uses the node ID selected during first-time setup:

```
Node ID: 2 through 13 (user selected)
```

Select **Save Configuration**. The Master saves the device entry and reboots automatically when required. The Periscope continues to report the Device ID saved during its own first-time setup.

Step 3 — Enter the Periscope Node ID and Master MAC

On first boot, the Periscope first asks for its node ID:

```
=====
FIRST TIME SETUP
=====

Enter Periscope Node ID (2-13):
```

Choose an unused ID from 2 through 13. The Master uses ID 0 and Displays use ID 1. Every DroidLink device must have a unique ID.

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After saving the node ID, the Periscope asks for the Master MAC:

```
Paste Master MAC like:  
XX:XX:XX:XX:XX:XX
```

Paste the DroidLink Master MAC address into the console.

The Periscope will confirm the value:

```
Received MAC: XX:XX:XX:XX:XX:XX
```

It will then save the MAC and reboot:

```
Master MAC saved!  
Rebooting...
```

The node ID and Master MAC are stored on the device and remain saved after power loss.

Successful Connection

When configuration is correct, the Periscope starts normally and appears in the Master's **Device Status** page. If its MAC has not yet been saved in a Master device slot, it may first appear as an **Unconfigured device**.

Add the displayed Periscope MAC to an available Master device slot and save the Master configuration to complete setup. When the Master sends a command, the Periscope console shows the received command and resulting lighting action:

```
:PSOFF  
LIGHT CMD: OFF
```

The Master and Periscope automatically find each other after both have been configured, regardless of which one powers on first.

Changing an Incorrect Master MAC

If the wrong Master MAC was entered, connect the Periscope to USB and open a serial terminal.

Type:

```
NEWMAC
```

The Periscope will erase the saved node ID and Master MAC:

```
Clearing stored Periscope configuration...  
Rebooting...
```

After reboot, the complete first-time setup returns. Enter the Periscope node ID first, followed by the correct Master MAC.

Paste Master MAC like:

XX:XX:XX:XX:XX

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Enter the correct Master MAC.

Master Communication Check

After boot, the Periscope checks for communication with the configured Master.

If Master communication is not detected within approximately 15 seconds, the console displays:

Master communication has not been detected yet.

If the saved Master MAC is incorrect, type NEWMAC to reconfigure.

This can mean:

- The Master is not powered on
- The wrong Master MAC was entered
- The Periscope MAC was not added to the Master
- The Master has not sent a command yet

DroidLink Command Format

All DroidLink Periscope commands begin with:

:PS

The complete format is:

:PS[PERISCOPE COMMAND]

Examples:

DroidLink Command	Periscope Receives	Description
:PSON	ON	Turn the lights on
:PSOFF	OFF	Turn the lights off
:PSX	X	Emergency off
:PSQ1	Q1	Party Mode
:PSQ4	Q4	Police Lights
:PSQ6	Q6	Knight Rider
:PSM185	M185	Main LEDs, white pulse
:PST647	T647	Top LEDs, blue chase
:PSA199	A199	All LEDs, pink fast

Basic Commands

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These are the native Periscope commands.

When using DroidLink, add `:PS` before the command.

Native Command	DroidLink Command	Description
ON	<code>:PSON</code>	Turn LEDs on
OFF	<code>:PSOFF</code>	Turn LEDs off
X	<code>:PSX</code>	Emergency off
?	<code>:PS?</code>	Show status/help when sent through Serial

Sequence Commands

A sequence command selects and configures the requested lighting sequence. If the Periscope LEDs are currently off, the sequence is prepared but will not be displayed until the lights are enabled with:

```
:PSON
```

For example, to select and display the Classic R2 Startup sequence:

```
:PSQ0  
:PSON
```

If the LEDs are already on, sending another sequence command changes to that sequence immediately. Use `:PSOFF` or `:PSX` to turn the LEDs off.

Native Command	DroidLink Command	Description
Q0	:PSQ0	Classic R2 Startup
Q1	:PSQ1	Party Mode
Q2	:PSQ2	Bright Pulse
Q3	:PSQ3	Communication Mode
Q4	:PSQ4	Police Lights
Q5	:PSQ5	Red Alert
Q6	:PSQ6	Knight Rider
Q7	:PSQ7	Searchlight
Q8	:PSQ8	Stealth Mode
Q9	:PSQ9	Diving
Q10	:PSQ10	Surfacing
Q11	:PSQ11	Calm Blue
Q12	:PSQ12	System Boot
Q13	:PSQ13	Fire Mode
Q14	:PSQ14	Celebration
Q15	:PSQ15	Charging
Q16	:PSQ16	Hyperdrive
Q17	:PSQ17	Malfunction
Q18	:PSQ18	Scan Complete
Q19	:PSQ19	Sonar Ping
Q20	:PSQ20	Auto Demo Mode

LED Groups

Code	Group
M	Main LEDs
T	Top LEDs
B	Bottom LEDs
S	Both sides
L	Left side
R	Right side
K	Back LEDs
A	All LEDs

Custom Command Format

The native Periscope format is: DroidLink Periscope Complete Guide · Document Revision 2 · September 26, 2026

```
[GROUP][EFFECT][COLOR][SPEED]
```

The DroidLink format is:

```
:PS[GROUP][EFFECT][COLOR][SPEED]
```

Example:

```
:PSM185
```

Meaning:

- :PS = Route to the Periscope Slave
- M = Main LEDs
- 1 = Effect 1
- 8 = White
- 5 = Medium speed

The Periscope receives:

```
M185
```

Colors

Number	Color
0	Red
1	Yellow
2	Green
3	Cyan
4	Blue
5	Magenta
6	Orange
7	Purple
8	White
9	Pink

Speed

Number	Speed
0	Slowest
5	Medium
9	Fastest

Values from 0 through 9 may be used.

Useful DroidLink Examples

Turn Everything Off

```
:PSOFF
```

Party Mode

```
:PSQ1
```

Police Lights

```
:PSQ4
```

Knight Rider

```
:PSQ6
```

Main LEDs — White Pulse

```
:PSM185
```

Top LEDs — Blue Chase

```
:PST647
```

All LEDs — Pink Fast

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```
:PSA199
```

Auto Demo

```
:PSQ20
```

Direct USB Serial Control

The Periscope can still be controlled directly through USB Serial.

When using Serial directly, do not add the DroidLink `:PS` prefix.

Use:

```
Q1
```

instead of:

```
:PSQ1
```

Use:

```
M185
```

instead of:

```
:PSM185
```

The main firmware serial handler also accepts:

```
NEWMAC
```

to erase the saved node ID and Master MAC and restart first-time setup.

Serial Settings

- Baud Rate: `9600`
- Line Ending: `Newline` or `Both NL & CR`

Commands are converted to uppercase by the firmware, so lowercase input is also accepted.

Hardware

Board

- ESP32-C3 Super Mini

LED Pins

Function	GPIO
Main	GPIO5
Right	GPIO7
Left	GPIO4
Bottom	GPIO3
Top	GPIO6
Back	GPIO10

Current DroidLink Configuration

The current firmware reports:

```
Node ID: configured value from 2 through 13
```

The Master routes commands beginning with:

```
:PS
```

to the configured Periscope.

Quick Setup Summary

1. Install Periscope firmware
2. Copy the Periscope MAC from the installer console
3. Add the Periscope MAC to the DroidLink Master
4. Select Save Configuration and allow the Master to reboot automatically when required
5. Enter an unused Periscope node ID from 2 through 13
6. Paste the Master MAC into the Periscope console
7. Periscope saves the node ID and Master MAC, then reboots
8. Confirm the Periscope appears in the Master's Device Status page
9. Send commands using :PS

Example:

```
:PSQ4
```

Runs Police Lights.

:PSOFF

Turns the Periscope lights off.

Backups and firmware updates

The Periscope does not provide a downloadable configuration backup. Record its Device ID, Master MAC, and device MAC somewhere safe so the setup can be restored if the controller is replaced or the flash is erased.

For a normal firmware update, use the DroidLink Installer with **Erase Flash disabled** so the saved setup is retained. If setup does not survive an update or you intentionally erase the flash, repeat the first-time setup in this guide.

Completion checklist

- ☐ Periscope firmware is installed
- ☐ Periscope is registered in the Master by its device MAC
- ☐ Periscope has an unused Device ID from 2 through 13
- ☐ Correct Master MAC is saved in the Periscope
- ☐ Master Device Status shows the Periscope connected
- ☐ :PS test commands operate the lights correctly
- ☐ Device ID and MAC addresses are recorded for recovery

Continue

Return to the [Documentation Home](#). The commands supported by the Periscope are documented in this guide.